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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
First Semester M.Tech. (VDE) (CBCS) Degree Examinations
January 2023

22VDE133 – SOC DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: Answer Five full questions choosing Two full questions from Unit – I & Unit – II each and One full question from Unit – III.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain Moore's law under the following heads.				
i)	Statement				
ii)	Lives and Deaths of Moore's law				
iii)	In recent years				
iv)	Future of Moore's law	10	L*2	1	3
b)	Explain any three Benefits of System-on-Chip in detail.	10	L2	1	3
2. a)	With a neat diagram and table compare SOB, SOC & SOP technologies.	10	L2	1	3
b)	Discuss the design productivity gap and solution for it.	10	L2	2	3
3. a)	Explain the SOC Intellectual Property (IP) in detail.	10	L2	2	3
b)	Write a note on Scaling. Show the result of technology scaling.	10	L2	2	3
Unit – II					
4. a)	With a neat diagram, explain the cache architecture of multi ported shared cache, private caches, and CCC.	10	L2	3	3
b)	How to reduce active and standby energy in case of energy-aware memory system design.	10	L3	3	3
5. a)	With a neat point to point diagram, explain one-directional bus connecting two modules.	10	L2	3	3
b)	Explain encoding for random white noise and correlated data.	10	L3	4	3
6. a)	What is packetization and energy efficiency? Explain energy and reliability in packet-switched NoCs.	10	L3	4	3
b)	Discuss the energy-aware software for MPSoCs.	10	L2	4	3
Unit – III					
7. a)	With a neat diagram explain HW/SW Co-design? What are the benefits of HW/SW co-design.	10	L2	5	3
b)	With a neat diagram explain, HW/SW Co-design at System-Level.	10	L2	5	3
8. a)	With a neat block schematic explain, HW/SW Co-design Flow. What are the Main phases of HW/SW co-design process?	10	L2	5	3
b)	With a neat architecture, explain VoIP SOC.	10	L2	5	3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
